

Introduction to Python

What is Python?

Python is a high-level, interpreted programming language designed with an emphasis on readability and simplicity. It was created by Guido van Rossum and first released in 1991. Python is now widely used in software development, automation, data science, artificial intelligence, machine learning, and backend development.

Why Learn Python?

Python has a simple syntax that is relatively easy for beginners to understand. It supports multiple programming styles, including procedural, object-oriented, and functional programming. Python also has a large ecosystem of libraries and frameworks, making it useful for both small scripts and large applications.

Common Data Types

Python includes built-in data types such as integers, floating-point numbers, strings, lists, tuples, dictionaries, sets, and Boolean values. These data types allow developers to represent and manipulate many kinds of information.

Applications

Python is commonly used with libraries such as NumPy and Pandas for data analysis, PyTorch and TensorFlow for machine learning, and FastAPI and Flask for backend development.